

- 3 (a)** Prove by mathematical induction that, for all positive integers n ,

$$\sum_{r=1}^n (5r^4 + r^2) = \frac{1}{2}n^2(n+1)^2(2n+1). \quad [6]$$

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

- (b) Use the result given in part (a) together with the List of formulae (MF19) to find $\sum_{r=1}^n r^4$ in terms of n , fully factorising your answer. [3]

[illegible]